SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I J.P. Mercy Lini (192521054) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: J.P.M.

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%	1	2	2	3
Method Selection	20%	1	2	2	2
Solution Process	25%	2	1	2	2
Accuracy of Calculation	20%	2	~~2~~ 3	1	1
Interpretation of Results	15%	2	1	1	1

Total Marks 34 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a /27 **subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

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1. Subnetting the network 192.168.20.0/24 for three labs

Given data

- Network address: 192.168.20.0/24
- Lab A requires a /26 subnet.
- Lab B requires a /27 subnet.
- Lab C requires a /28 subnet.

Formulas used

1. Number of addresses per subnet = $2^{32-\text{prefix length}}$
2. Usable host addresses = $2^{32-\text{prefix length}}-2$
3. Block size = $256-$subnet mask's last octet

The first and last addresses of each IPv4 subnet are reserved for the network and broadcast addresses, respectively.

Lab A — /26

Subnet mask: 255.255.255.192

- Block size = $256-192=64$
- Total addresses = $2^{32-26}=64$
- Usable hosts = $64-2=62$

Allocation:

- Network address: 192.168.20.0
- First valid host: 192.168.20.1
- Last valid host: 192.168.20.62
- Broadcast address: 192.168.20.63

Lab B — /27

Subnet mask: 255.255.255.224

- Block size = $256-224=32$
- Total addresses = $2^{32-27}=32$
- Usable hosts = $32-2=30$

Assuming Lab A is allocated first, Lab B can use the next non-overlapping subnet.

- Network address: 192.168.20.64
- First valid host: 192.168.20.65
- Last valid host: 192.168.20.94
- Broadcast address: 192.168.20.95

Lab C — /28

Subnet mask: 255.255.255.240

- Block size = $256 - 240 = 16$
- Total addresses = $2^{32-28} = 16$
- Usable hosts = $16 - 2 = 14$

Assuming Lab A and Lab B are allocated first:

- Network address: 192.168.20.96
- First valid host: 192.168.20.97
- Last valid host: 192.168.20.110
- Broadcast address: 192.168.20.111

Final answer

Details	**Lab A**	**Lab B**	**Lab C**
CIDR prefix	/26	/27	/28
Subnet mask	255.255.255.192	255.255.255.224	255.255.255.240
Network address	192.168.20.0	192.168.20.64	192.168.20.96
Valid host range	.1 – .62	.65 – .94	.97 – .110
Broadcast address	.63	.95	.111
Total addresses	64	32	16
Usable hosts	62	30	14

Interpretation: The three labs receive separate, non-overlapping subnets. A total of $62+30+14=106$usable host addresses are available. The remaining range, 192.168.20.112–192.168.20.255, is available for future allocation, subject to the chosen subnet boundaries.

2. Subnet analysis of two systems using a /26 **mask**

Given data

- System 1 IP: 192.168.1.130
- System 2 IP: 192.168.1.190
- Subnet mask: 255.255.255.192 (/26)

Step 1: Calculate the block size

The last octet of the subnet mask is 192.

$$\text{Block size} = 256 - 192 = 64$$

Therefore, the subnet ranges in the last octet are:

- 0–63
- 64–127
- 128–191
- 192–255

Step 2: Identify the subnet for each system

Both 130 and 190 fall within the range 128–191. Hence, both systems belong to the same /26 subnet.

Network address: 192.168.1.128

Broadcast address: 192.168.1.191

Valid host range: 192.168.1.129–192.168.1.190

Step 3: Calculate usable host addresses

$$\text{Total addresses} \wedge ¿ 2^{32-26}$$
$$¿ = 2^{6} = 64$$
$$\text{Usable hosts} \wedge ¿ 64 - 2$$
$$¿ = 62$$

Final answer

Details	**System 1**	**System 2**
IP address	192.168.1.130	192.168.1.190
Subnet mask	/26	/26
Network address	192.168.1.128	192.168.1.128
Valid host range	192.168.1.129–190	192.168.1.129–190
Broadcast address	192.168.1.191	192.168.1.191
Usable hosts	62	62
IP validity	Valid host	Valid host

Interpretation: Both IP addresses belong to the same subnet, 192.168.1.128/26. Both are valid host addresses, and the subnet provides 62 usable host addresses. System 2's address, .190, is the last valid host address before the broadcast address .191.

3. Subnet allocation for HR, Sales, IT, and Admin

Given data

The organization has the network block 172.16.0.0/16.

Department	Required prefix
HR	/25
Sales	/26
IT	/27
Admin	/28

We allocate the subnets consecutively, starting from 172.16.0.0, ensuring that each subnet begins at a valid network boundary and does not overlap another.

Step 1: HR department — /25

- Subnet mask: 255.255.255.128
- Total addresses: $2^{32-25}=128$
- Usable hosts: $128-2=126$
- Block size: $256-128=128$

Allocation:

- Network address: 172.16.0.0
- Valid host range: 172.16.0.1–172.16.0.126
- Broadcast address: 172.16.0.127

Step 2: Sales department — /26

- Subnet mask: 255.255.255.192
- Total addresses: $2^{32-26}=64$
- Usable hosts: $64-2=62$
- Block size: $256-192=64$

The next available address is 172.16.0.128, which is a valid /26 boundary.

Allocation:

- Network address: 172.16.0.128
- Valid host range: 172.16.0.129–172.16.0.190
- Broadcast address: 172.16.0.191

Step 3: IT department — /27

- Subnet mask: 255.255.255.224
- Total addresses: $2^{32-27}=32$

- Usable hosts: $32-2=30$
- Block size: $256-224=32$

The next available address is 172.16.0.192.

Allocation:

- Network address: 172.16.0.192
- Valid host range: 172.16.0.193–172.16.0.222
- Broadcast address: 172.16.0.223

Step 4: Admin department — /28

- Subnet mask: 255.255.255.240
- Total addresses: $2^{32-28}=16$
- Usable hosts: $16-2=14$
- Block size: $256-240=16$

The next available address is 172.16.0.224.

Allocation:

- Network address: 172.16.0.224
- Valid host range: 172.16.0.225–172.16.0.238
- Broadcast address: 172.16.0.239

Final allocation table

Department	Network address	Valid host range	Broadcast address	Usable hosts
HR	172.16.0.0/25	172.16.0.1–126	172.16.0.127	126
Sales	172.16.0.128/26	172.16.0.129–190	172.16.0.191	62
IT	172.16.0.192/27	172.16.0.193–222	172.16.0.223	30
Admin	172.16.0.224/28	172.16.0.225–238	172.16.0.239	14

Step 5: Calculate the total allocated addresses

$$128+64+32+16=240$$

Total usable host addresses across the four departments:

$$126+62+30+14=232$$

The original /16 network contains:

$$2^{32-16}=65{,}536 \text{ addresses}$$

Therefore, the number of addresses not included in the four allocated subnets is:

$$65{,}536 - 240 = 65{,}296$$

Remaining unused IP address range

The four subnets occupy 172.16.0.0–172.16.0.239.

The remaining unused address range is:

172.16.0.240–172.16.255.255

This is a continuous range of unallocated IP addresses; it can be divided into additional subnets as required.

Interpretation: Each department receives a non-overlapping subnet with a different capacity. HR has the largest allocation, while Admin has the smallest. A total of 232 usable host addresses are provided by the current allocations, and 65,296 addresses remain outside those allocated subnets.

4. Subnetting 172.16.10.0/24 **using a** /27 **mask**

Given data

- Original network: 172.16.10.0/24
- New subnet mask: /27
- IP address to analyze: 172.16.10.78
- IP address to check: 172.16.10.95

Step 1: Determine the number of subnets created

The original prefix is /24, and the new prefix is /27.

Number of borrowed bits:

$$27 - 24 = 3$$

Number of subnets:

$$2^3 = 8 \text{ subnets}$$

Step 2: Calculate the usable hosts per subnet

The /27 subnet mask is 255.255.255.224.

Host bits:

$$32 - 27 = 5$$

Total addresses per subnet:

$$2^5 = 32$$

Usable host addresses:

$$2^5 - 2 = 32 - 2 = 30$$

Step 3: Determine the subnet containing 172.16.10.78

Block size:

$$256 - 224 = 32$$

The eight subnet ranges are:

Subnet number	Network address	Broadcast address
1	172.16.10.0	172.16.10.31
2	172.16.10.32	172.16.10.63
3	172.16.10.64	172.16.10.95
4	172.16.10.96	172.16.10.127
5	172.16.10.128	172.16.10.159
6	172.16.10.160	172.16.10.191
7	172.16.10.192	172.16.10.223
8	172.16.10.224	172.16.10.255

The address 172.16.10.78 falls within the third subnet, 172.16.10.64–172.16.10.95.

Therefore:

- Network address: 172.16.10.64
- First valid host: 172.16.10.65
- Last valid host: 172.16.10.94
- Broadcast address: 172.16.10.95

Step 4: Check 172.16.10.95

The address 172.16.10.95 is the broadcast address of the third subnet.

Therefore, it belongs to the same subnet's address range, but **it cannot be assigned to an individual host** because it is reserved for broadcasting.

Final answer

Parameter	Result
Original network	172.16.10.0/24
New subnet mask	255.255.255.224 (/27)
Number of subnets	8

Total addresses per subnet	32
Usable hosts per subnet	30
Subnet of 172.16.10.78	172.16.10.64/27
Network address	172.16.10.64
Valid host range	172.16.10.65–172.16.10.94
Broadcast address	172.16.10.95
Is .78 a valid host?	Yes
Is .95 in the same subnet?	Yes, as the broadcast address; not a valid host

Interpretation: Dividing the /24 network into /27 subnets creates eight subnets with 30 usable host addresses each. The address .78 is a valid host in the third subnet, whereas .95 is its broadcast address.

5. IPv6 address compression, expansion, and /64 **prefix**

Given data

IPv6 address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Prefix length: /64

An IPv6 address contains 128 bits divided into eight groups of 16 bits, called hextets. Each hextet is represented by four hexadecimal digits in its full notation.

Step 1: Compress the IPv6 address

IPv6 compression follows two rules:

1. Leading zeros in each hextet can be removed.
2. A consecutive sequence of all-zero hextets can be replaced by :: once in an address.

Original address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Remove leading zeros from each hextet:

2001:db8:0:0:0:ff00:42:8329

Replace the longest consecutive sequence of zero hextets with ::.

Shortest form:

2001:db8::ff00:42:8329

Step 2: Expand the compressed address

The compressed address is:

2001:db8::ff00:42:8329

It contains five explicitly written hextets, so the :: represents three zero hextets to make eight hextets in total.

Expanded form:

2001:0db8:0000:0000:0000:ff00:0042:8329

This matches the original IPv6 address.

Step 3: Determine the network prefix using /64

A /64 prefix means that the first 64 bits (the first four hextets) represent the network portion.

The first four hextets are:

- 2001
- 0db8
- 0000
- 0000

Therefore, the network prefix is:

2001:db8:0:0::/64

The remaining 64 bits are available for interface identifiers within this network.

Step 4: Calculate the number of possible host addresses

Total address bits = 128

Network bits = 64

Host bits:

$$128 - 64 = 64$$

Number of possible interface identifiers:

$$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$$

Thus, the /64 network contains 2^{64} possible interface identifier values. Unlike conventional IPv4 subnet calculations, IPv6 does not subtract two addresses for network and broadcast addresses; IPv6 does not use broadcast addressing.

Final answer

Parameter	**Result**
Original IPv6 address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed address	2001:db8::ff00:42:8329

Expanded address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix length	/64
Network prefix	2001:db8:0:0::/64
Network bits	64
Interface identifier bits	64
Possible interface identifier values	2^{64}
Total possible values	18,446,744,073,709,551,616

Interpretation: IPv6 compression shortens the address without changing its meaning. A /64 prefix divides the address into a 64-bit network portion and a 64-bit interface identifier portion, providing an enormous address space for devices on the network.

Overall conclusion

Subnetting divides a larger IP network into smaller, manageable networks. The subnet mask determines each subnet's size, network address, valid host range, and broadcast address in IPv4. IPv6 uses a 128-bit addressing system and supports compression to simplify address representation. Correct subnet allocation prevents overlap and enables efficient use of network resources.

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